

FINAL EXPLANATION: BIOLOGY GRADE 11

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 11: REPRODUCTION IN ANIMALS (SUB-STRAND 3.1) – FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving question: How do different animals ensure successful fertilisation and development of offspring, and how can science assist when human reproduction fails? Use what you observed and learned in Lessons 1 to 7 to write your Final Explanation. Answer each prompt in the space below it, using full sentences and the correct scientific words. Refer to the frogs near Kisumu and the Lake Victoria wetlands, the hen, the cow and the IVF stories you met in Lesson 1. Give evidence from your class work, such as the dissection or annotated photographs, the sperm and ovum slides or charts, the hormone graphs, the animations, the case studies and the debate. Your answers will also form part of your portfolio or poster and your position paper on reproductive biotechnology.

Part 1: Frogs, Hens and Cows – Many Ways to One Goal

Frogs near Kisumu release thousands of jelly-covered eggs into water. A hen lays hard-shelled eggs on land. A cow gives birth to one live calf. (a) Explain where fertilisation takes place in the frog, the hen and the cow, and name the type of fertilisation in each. (b) Use the terms oviparous and viviparous to describe how the young of the frog, hen and cow develop. Then explain what ovoviviparous means, using an example from your classification charts. (c) Explain why the frog needs thousands of eggs but the cow produces one calf. Think about the site of fertilisation, losses of eggs and young, and parental care.

Write your answer here:

Part 2: Built for Reproduction – Structure and Function

(a) From your dissection of a small mammal (or the annotated photographs, models and charts), name the parts of the male system that a sperm passes through, and the main parts of the female system. (b) State what the testes and ovaries do, where fertilisation occurs, and the role of the uterus. (c) Describe the structure of a sperm cell and of an ovum, and explain how each structure helps the cell to do its job (movement, protection or fusion).

Write your answer here:

Part 3: Hormonal Conductors – Cycle, Pregnancy and Birth

(a) Describe how FSH, LH, oestrogen and progesterone change during a 28-day menstrual cycle and what each does. Include what happens near Day 14. (b) Explain how feedback between the pituitary gland and the ovaries helps control the timing of egg release, and compare this with the frog's release of eggs. (c) Explain how hormones and the placenta support the baby if fertilisation occurs, from implantation to birth and lactation. Include hCG, progesterone, oxytocin and prolactin. (d) Suggest what could happen if a hormone is undersecreted or oversecreted.

Write your answer here:

Part 4: When Reproduction Fails – IVF, Surrogacy and Sperm Banking

(a) Many people think that IVF babies grow wholly in a test tube. Use what you learned to decide whether this is correct and explain what happens in IVF and where the baby develops. (b) For each of the couples in the case studies, say which technology (IVF, surrogacy or sperm banking) could help and why: blocked fallopian tubes (Nakuru), low sperm count (Kisumu), hormonal imbalance (Eldoret), sperm storage before cancer treatment (Nairobi), absent uterus (Mombasa). (c) Describe one ethical issue that must be considered.

Write your answer here:

Part 5: Putting It All Together

Write a final answer of about 8 to 10 sentences to the driving question: How do different animals ensure successful fertilisation and development of offspring, and how can science assist when human reproduction fails? Include (1) external and internal fertilisation, with the frog, hen and cow as examples, (2) the role of structure and hormones in humans, (3) the role of biotechnology, and (4) how your thinking has changed since Lesson 1.

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Comparing animal strategies	Accurately explains external and internal fertilisation and the oviparous, viviparous and ovoviviparous types with the frog, hen and cow, and links the number of offspring to the site of fertilisation, losses and parental care.	Correctly describes the types of fertilisation and development with examples, but the link to the number of offspring is only partly explained.	Names some types or examples but with errors, or does not explain why frogs need thousands of eggs and a cow only one calf.
Structure and function	Correctly names the reproductive parts and explains how the sperm, ovum, oviduct and uterus are adapted to their functions, using evidence from the dissection and slides.	Names most parts and describes functions correctly, with limited explanation of how structure suits function.	Names few parts or confuses their functions.
Hormonal control	Clearly explains the roles and timing of FSH, LH, oestrogen and progesterone, feedback loops, and the roles of hCG, the placenta, oxytocin and prolactin, including the effects of imbalance.	Describes most hormones and their roles correctly but does not fully explain feedback or the pregnancy stages.	Mentions hormones without correct roles or sequence.
Biotechnology and ethics	Matches IVF, surrogacy and sperm banking to the specific causes of infertility, corrects the test-tube misconception and discusses ethical issues with reasoning.	Explains the technologies correctly and mentions one ethical issue with little reasoning.	Describes technologies inaccurately or keeps the test-tube misconception.
Communication and use of evidence	Writes a clear, well-organised answer using correct scientific terms and evidence from the lessons, and shows how thinking changed since Lesson 1.	Writes a mostly clear answer with some scientific terms and some evidence.	Writes a brief or unclear answer with few scientific terms and little evidence.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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